

atomOS

Misuse threat atlas

A first-principles map of computational, spatial and physical possibility spaces

40 SCENARIOS / 7 COLORS / 7 MUSICAL NOTES

L1 / C	L2 / D	L3 / E	L4 / F	L5 / G	L6 / A	L7 / B
Trace	Limited	Material	Serious	Severe	Critical	Extreme

The central finding

The formalization describes ways to encode, transform and verify finite computational states. Misuse becomes concrete when an actor controls inputs, program publication, claims made about results, or an added interface to people and physical systems.

Read the colors as conditional harm

The flags describe consequences **if the stated prerequisites are supplied**. They do not measure likelihood or identify an active threat. No malicious deployment or external physical connection is established by the reviewed evidence.

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Formalization and required log-polar/Klein direction: Tom Klootwijk
Threat analysis, invented scale and layout: AI-assisted synthesis
Version 1.0 | 14 September 2026

What exists, and what follows

A source-grounded scope statement before assigning a threat color.

The implemented computational core

The required format combines seeded, packed one-bit program pages on a log-polar chart with Klein seam handling and a fixed source-word SDF NOR operator atlas. The current bank has seven pages, including three admitted teacher-derived procedures: a full adder, binary-to-Gray conversion and a selector. [K1, K2]

Measured texture residency has a bounded scope

The documented program pages plus operator atlas occupy **2,816 texture bytes**. Four cold profiles recorded the expected 88 compulsory 32-byte sector misses and no additional misses. This supports retention within those measured launches. Permanent cache pinning, arbitrary-workload residency and general performance superiority are unproved. [K2]

The current knowledge route

A local teacher pipeline uses a loopback Ollama API and cached models to propose finite algorithms. Admission checks the unchanged finite specification and rejects duplicate functions. The admitted bank holds executable procedures; it does not hold the teachers' weights. General language behavior and a persistent native texture worker between queries remain unfinished. [K2]

The physical boundary

The reviewed core has no demonstrated live sensor feed, external identity matcher, neural or biochemical interface, physical actuator or weapon controller. The main formalization itself identifies missing interfaces and rejects treating named external systems as established connections. Source language about such systems is not evidence of access to them. [M1 pp. 24, 29, 36, 40]

Correctness does not settle purpose

A computation can faithfully implement a harmful specification. A finite truth-table check does not prove empirical truth, consent, safe deployment or general harmlessness. Likewise, semantic novelty does not by itself establish usefulness. This distinction drives the entire atlas. [K1; R1, R2]

Where the misuse could attach

Primitive-level reading of the main formalization. Page numbers refer to physical PDF pages in M1.

Primitive / source	What the formal object does	Boundary relevant to misuse
SHA-256 identity and seed p. 8	Deterministically projects canonical records into chart seeds; full identity stays separate.	No consent, authentication, external location measurement or account connection follows.
Log-polar chart p. 9	Represents finite cells through log radius, periodic angle and declared quantization.	Physical use needs measurements; chart indices and physical units remain distinct.
Mandatory Klein seam p. 10	Reflects radial position at angular winding in the required current format.	Coordinate identification creates no physical transport, remote access or device coupling.
Word operators and lineage p. 11, 14-17	Defines distinct shifts, lineage, JK state, parity and finite word rules.	Recurrence is not external replication; parity is not identity or authorization.
SDF words and absorption p. 16-18	Applies supplied predicates and ordered ASA/NA rules, including whole-word zeroing.	Binary word rules do not establish physical absorption or every scalar geometry.
OTAN2 and frames p. 19	Uses literal atan of the increment ratio and an explicitly supplied axis.	Frames, increments and valid statuses remain required; no tracker appears automatically.
Occupancy versus identity p. 24	Quantizes declared candidates while retaining separate identity and lineage records.	Shared occupancy cannot identify each record or create sensors and actuators.
Snapshots and replay p. 25, 30	Commits verified snapshots and links events for conditionally deterministic replay.	Replay proves neither authority nor truth; rollback cannot undo physical effects.
CUDA and texture storage p. 27-28	Preserves logical cells through packed layouts and immutable texture reads.	No host privilege, encryption or permanent cache pinning follows from this layout.
Computational universality p. 31-32	Simulates finite machine prefixes when definitions, storage and execution budget suffice.	No infinite memory, general intelligence, arbitrary termination or binary proof follows.

Current design requirement: the Klein topology is part of the requested substrate. This atlas does not recast it as an optional feature. Packing and topology change representation and adjacency; any physical application still needs its own justified coupling.

The rainbow and tone legend

An original ordinal scale for **conditional consequence severity**, created for this atlas. It is not an official standard or a probability model.

L1 / C Trace Violet	Minor nuisance or readily reversible local disruption; no sensitive data or bodily harm in the stated scenario.
L2 / D Limited Indigo	Recoverable local loss or unfairness affecting a limited activity, with a practical route to recovery.
L3 / E Material Blue	Meaningful financial, privacy or operational harm, or substantial waste of resources.
L4 / F Serious Green	Substantial individual harm, coercion, discrimination or a persistent violation of rights.
L5 / G Severe Yellow	Serious injury or major consequences for a community, organization or service.
L6 / A Critical Orange	Plausible loss of life, multiple serious injuries or sustained disruption of an essential service.
L7 / B Extreme Red	Catastrophic mass harm or irreversible systemic damage, conditional on independently capable external systems.

How to interpret a flag

A flag rates the scenario as described, assuming its prerequisites. The bands are ordered judgments with overlapping real-world consequences; they are not equal numerical steps. A different scale of deployment can justify a different band. Likelihood is **unassessed**: there is no actor, exposure or incident dataset here. [R1]

Accessibility and meaning: the number, note and name repeat every color. Green means L4 / F / Serious here, not "safe". Notes C-D-E-F-G-A-B are memory aids only, with no acoustic, physiological or resonance prescription. Text labels apply the principle of not relying on color alone. No WCAG or PDF/UA conformance claim is made. [R8]

Evidence and the physical bridge

Use the path flag alongside the harm color. An existing component is not proof that its misuse succeeds.

K / Kernel-adjacent

A relevant computational component exists in the reviewed package. The stated abuse has not been demonstrated. An actor would still need the specified ability to supply input, replace data or influence publication.

A / Added application

The scenario requires application logic, sensitive data, a communication channel or permission not established by the core. Existing technology elsewhere does not establish this repository's access to it.

P / Physical interface

The scenario additionally requires real sensors, effectors or material processes, actual physical coupling and relevant access. This is a dependency label, not a claim that every named physical technology is scientifically speculative. Authorization and validation are safeguards, not prerequisites for causing harm.

THE REQUIRED CAUSAL CHAIN



U / Grey: an unsupported causal leap

No operational threat level is assigned when the proposed mechanism has no established bridge. Examples include a hash conferring external authority, a Klein seam physically connecting remote locations, or word absorption by itself removing physical energy. Claims of infinite hardware or established neural/weapon access likewise exceed the evidence. Grey means unsupported, not safe. [M1 pp. 8, 10, 18, 31-32, 40]

There is no demonstrated atomOS-specific advantage for misuse. Speed or compactness might change the scale of a real application, but that would require comparative measurements and a deployment analysis.

Scenario index / 1

Scan the consequence flag first, then check the required path. K = kernel-adjacent; A = added application; P = physical interface.

ID / scenario	Level / note	Path	Page
T01 Local demo sabotage	 L1 / C	K/A	8
T02 Game cheating and unfair simulation	 L2 / D	A	8
T03 Unauthorized program-bank substitution	 L4 / F	K	8
T04 Compiler or parser input abuse	 L3 / E	K	8
T05 Oracle or verifier capture	 L5 / G	K	9
T06 Unauthorized rollback or mixed versions	 L4 / F	K/A	9
T07 False novelty through repetition	 L3 / E	K	9
T08 Hash or provenance impersonation	 L4 / F	K	9
T09 Unproductive self-reference	 L3 / E	K	10
T10 Resource denial and monopolization	 L3 / E	K/A	10
T11 Shared-GPU information leakage	 L4 / F	A	10
T12 Logs and outputs as disclosure channels	 L4 / F	A	10
T13 Restricted knowledge extraction	 L4 / F	A	11
T14 Exact amplification of false premises	 L4 / F	K/A	11
T15 Fabricated research or benchmark evidence	 L4 / F	K/A	11
T16 Universality, residency or AI claim fraud	 L3 / E	K	11
T17 Identity and occupancy confusion	 L4 / F	A	12
T18 Frame, unit or quantization deception	 L5 / G	A/P	12
T19 Sensor or input poisoning	 L5 / G	A/P	12
T20 Optical, RF or acoustic device misuse	 L6 / A	P	12

A severe color is not a current capability finding

Orange and red entries depend on independently capable external systems. Their inclusion anticipates governance needs if such systems are ever connected. No current connection or malicious activity is inferred from the formalization.

Scenario index / 2

Scan the consequence flag first, then check the required path. K = kernel-adjacent; A = added application; P = physical interface.

ID / scenario	Level / note	Path	Page
T21 Persistent person tracking	 L4 / F	A	13
T22 Private XR-map abuse	 L4 / F	A	13
T23 Biometric or emotional inference coercion	 L4 / F	A/P	13
T24 Automated discrimination and exclusion	 L4 / F	A	13
T25 Impersonation and deceptive interaction	 L4 / F	A	14
T26 Exploitative adaptive engagement	 L4 / F	A	14
T27 Unsafe XR or sensory output	 L5 / G	A/P	14
T28 Nonconsensual intimate simulation	 L4 / F	A/P	14
T29 Neural privacy and false mind claims	 L4 / F	A/P	15
T30 Unsafe neural modulation	 L6 / A	P	15
T31 Biochemical or physiological misuse	 L6 / A	P	15
T32 Unvalidated clinical claims	 L5 / G	A/P	15
T33 Unsafe robotics or machinery	 L6 / A	P	16
T34 Critical-service or OT sabotage	 L6 / A	P	16
T35 Hazardous navigation	 L5 / G	P	16
T36 Weapon-system adaptation	 L7 / B	P	16
T37 Hazardous science enablement	 L7 / B	A/P	17
T38 Harmful environmental intervention	 L5 / G	A/P	17
T39 Institutional gatekeeping and lock-in	 L4 / F	A	17
T40 Energy and economic externalities	 L3 / E	A	17

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D1 - Trust and entry points

Who can supply inputs, replace content or claim authority determines the entry points for misuse, from reversible nuisance to consequential unauthorized changes.

T01 Local demo sabotage

L1 / C / TRACE

Path and harm: Access to a disposable local demo could permit misleading display or fixture changes, causing confusion and wasted attention. This level assumes no sensitive data, persistent assets, external decisions or physical outputs.

Warning sign: Displayed behavior differs from the retained synthetic fixture.

Safeguard: Keep demos isolated and resettable; label synthetic inputs, retain reference outputs, and compare fixtures before presentation.

Path K/A | K1; R1

T02 Game cheating and unfair simulation

L2 / D / LIMITED

Path and harm: A game trusting client-generated rules or outcomes could permit unauthorized advantages, altered scores and unfair play. This level assumes recoverable recreational harm.

Warning sign: Competitive results depend on unchecked local banks or unexplained rule changes.

Safeguard: Validate authoritative outcomes independently, identify the active rule version, separate experiments from competition, and provide replayable disputes and restoration.

Path A | R1; R2

T03 Unauthorized program-bank substitution

L4 / F / SERIOUS

Path and harm: Replacing an accepted bank or manifest could redirect significant downstream decisions to an unauthorized function. Level 4 assumes consequential use, not a harmless demo.

Warning sign: Bank changes lack an approved publication receipt, or provenance and active versions disagree.

Safeguard: Authenticate publishers, bind publication to checked content, verify before loading, preserve a known-good epoch, and record accountable changes.

Path K | K1; K2; R1

T04 Compiler or parser input abuse

L3 / E / MATERIAL

Path and harm: Malformed replies or packed content reaching a vulnerable parser could cause disruption, unintended interpretation or local compromise. No current exploit is asserted.

Warning sign: Inputs exceed declared limits, parsing becomes inconsistent, or processing touches unexpected files.

Safeguard: Use declarative formats, bounded parsing and isolated compilation; reject executable input, retain malformed regression samples, and stop publication on validation failure.

Path K | K1; K2; R1

D2 - Validation and provenance

Trustworthy execution depends on trustworthy specifications, evidence and publication authority. Hashes and successful checks cannot repair a corrupted source of expected behavior.

T05 Oracle or verifier capture

L5 / G / SEVERE

Path and harm: Control over expected results, checking logic or evidence could give wrong or harmful programs a verified label. Severe harm requires consequential downstream reliance.

Warning sign: Candidate and expected outputs change together, failures disappear, or independent specifications vanish.

Safeguard: Separate proposal, specification and publication authority; freeze originals, independently reproduce acceptance, review verifier changes, and reject self-comparison as external confirmation.

Path K | K1; K2; R1

T06 Unauthorized rollback or mixed versions

L4 / F / SERIOUS

Path and harm: Unauthorized rollback or inconsistent publication could restore withdrawn behavior or mix incompatible state. Level 4 assumes significant downstream consequences rather than a disposable demonstration.

Warning sign: Versions reverse without explanation, lineage breaks, or state and program epochs disagree.

Safeguard: Require authorized transitions and compatible versions; publish atomically after checks, record complete receipts, and preserve recoverable versions still eligible for use.

Path K/A | K1; K2; R1

T07 False novelty through repetition

L3 / E / MATERIAL

Path and harm: Counting new seeds, names or equivalent circuits as knowledge could inflate the bank, waste review resources and misrepresent progress.

Warning sign: Pages and lineage grow without new useful input-output behavior.

Safeguard: Compare semantics within the declared domain, preserve interface meaning, require independent correctness and usefulness, cap acquisition work, and record duplicate rejection without adding an active page.

Path K | K2; R1; R2

T08 Hash or provenance impersonation

L4 / F / SERIOUS

Path and harm: Presenting a digest or lineage as proof of truth, consent or authorship could give false claims authority. Hashes bind bytes, not those conclusions.

Warning sign: Hash-verified claims lack authenticated origin, factual support or relevant consent.

Safeguard: Explain each digest's scope, authenticate publishers separately, retain source context and permissions, and require independent evidence for substantive claims.

Path K | K1; K2; R1

D3 - Runtime and data boundaries

Bounded progress, isolation and output controls complement resource limits and immutable texture reads, especially when applications process confidential information or share hardware.

T09 Unproductive self-reference

L3 / E / MATERIAL

Path and harm: Unbounded recurrence could exhaust resources or recycle a program's outputs into supposed evidence.

Warning sign: States and claims recur, descendants cite only themselves, and work grows without validated gains.

Safeguard: Set time, memory and proposal budgets; detect repeated states, require independent evidence, preserve legitimate bounded recurrence, and stop acquisition when useful progress cannot be demonstrated.

Path K | K1; K2; R1

T10 Resource denial and monopolization

L3 / E / MATERIAL

Path and harm: Excessive allocation or scheduling privileges could exhaust VRAM, compute or verification capacity and interrupt other work.

Warning sign: Memory pressure persists, unrelated jobs starve, or admission repeatedly fails with little useful output.

Safeguard: Enforce quotas, fair scheduling, cancellation and reserves; account for acquisition and verification costs, and preserve resources for the user and recovery processes.

Path K/A | K1; R1

T11 Shared-GPU information leakage

L4 / F / SERIOUS

Path and harm: Shared hardware could expose confidential information if an adversarial observer can exploit a real leakage mechanism. Cache measurements alone demonstrate no such extraction.

Warning sign: Sensitive tenants share resources against policy, or controlled assessment finds reproducible information leakage.

Safeguard: Assess the actual platform, isolate sensitive workloads appropriately, minimize secret-dependent behavior, and retain supported isolation controls.

Path A | K1; R1

T12 Logs and outputs as disclosure channels

L4 / F / SERIOUS

Path and harm: An application handling confidential inputs could disclose them through exported traces, logs, receipts or answers despite correct computation.

Warning sign: Private inputs appear in public evidence, unnecessary logs persist, or destinations exceed the approved workflow.

Safeguard: Minimize sensitive retention, separate public receipts from protected evidence, control exports and destinations, and preserve sufficient restricted information for an independent audit.

Path A | K1; K2; R1

D4 - Knowledge and evidence

Deterministic procedures can carry false assumptions, restricted information or overstated claims. Exact computation does not establish truth, scope or legitimate use.

T13 Restricted knowledge extraction

L4 / F / SERIOUS

Path and harm: Acquisition accessing confidential, personal or restricted sources could reproduce information without permission. Repacking does not remove confidentiality or source obligations.

Warning sign: Admitted content contains secrets or identifiable records with unclear provenance.

Safeguard: Use authorized sources, retain acquisition provenance, review disclosure before publication, restrict sensitive packs, and support tracing, withdrawal and replacement when source permissions fail.

Path A | K2; R1; R2

T14 Exact amplification of false premises

L4 / F / SERIOUS

Path and harm: Treating empirical or conditional assertions as universally true could produce repeatable wrong decisions.

Warning sign: Rules omit scope, dates or uncertainty, discard contradictions, or answer outside their validated applicability.

Safeguard: Represent scope and dependencies, preserve unknown and conflict states, review changing claims, and require independent domain evidence rather than treating teacher agreement or finite procedural correctness as general truth.

Path K/A | K2; R1; R2

T15 Fabricated research or benchmark evidence

L4 / F / SERIOUS

Path and harm: Synthetic traces, selective tests or generated observations presented as measurements could mislead research and downstream decisions.

Warning sign: Raw data are absent, failures disappear, synthetic inputs become reported experiments, or thresholds change after failure.

Safeguard: Retain originals and failures, label synthetic and not-run evidence, bind results to exact sources and binaries, disclose exclusions, and enable independent reproduction.

Path K/A | K1; K2; R1

T16 Universality, residency or AI claim fraud

L3 / E / MATERIAL

Path and harm: Marketing finite tests as unlimited intelligence, permanent cache pinning or established physical control could induce costly reliance through deception.

Warning sign: Claims jump from NOR completeness or cache counters to untested abilities and guarantees.

Safeguard: Maintain a claim-to-evidence ledger; distinguish assumptions, proofs, native execution and measurements, publish limitations beside results, and require application-specific evidence before expanding claims.

Path K | M1 p. 32; K1; K2; R1

D5 - Geometry and sensing

Coordinates and predicates require explicit meaning. Physical consequences need additional sensing, physical coupling or output hardware connecting representations to the world.

T17 Identity and occupancy confusion

L4 / F / SERIOUS

Path and harm: Confusing cell predicates, blocked-region bits, object identifiers and personal identity could cause false matches or consequential misclassification.

Warning sign: Data cross incompatible dictionaries, or sampled membership is presented as proof of identity.

Safeguard: Type representations, bind dictionaries and sampling conventions, verify conversions, preserve missing classifications, and require independent identification evidence before attributing geometric results to a person or object.

Path A | M1 pp. 8, 16-17, 24; K1; R1

T18 Frame, unit or quantization deception

L5 / G / SEVERE

Path and harm: Concealed changes to frames, units, reference lengths or quantization could corrupt geometry and cause serious harm where physical decisions rely on it.

Warning sign: Units or frame identities disappear, seams disagree, or precision claims exceed resolution.

Safeguard: Bind units, frames and uncertainty to data; validate conversions and boundaries, preserve undefined states, and independently calibrate safety-relevant physical use.

Path A/P | M1 pp. 9-10; K1; R3

T19 Sensor or input poisoning

L5 / G / SEVERE

Path and harm: False or altered observations entering a connected sensing pipeline could drive dangerous conclusions despite correct LUT evaluation.

Warning sign: Independent observations conflict, calibration changes lack explanation, or implausible measurements receive certainty.

Safeguard: Protect data paths, cross-check independent sources, monitor quality and calibration, retain uncertainty, and ensure the application responds safely when inputs violate its operating assumptions.

Path A/P | K1; R1; R3

T20 Optical, RF or acoustic device misuse

L6 / A / CRITICAL

Path and harm: A coupled optical, RF or acoustic device could cause injury, interference or unauthorized sensing. Level 6 applies only where death or multiple severe injuries are credible; otherwise rate the stated harm.

Warning sign: Outputs exceed the approved purpose, safeguards are bypassed, or calibration is unsupported.

Safeguard: Enforce independent device limits, authorize outputs, validate the apparatus, monitor effects, and stop on uncertainty.

Path P | M1 p. 18; K1; R3

D6 - Privacy and social decisions

Personal data and decision authority create risks absent from synthetic fixtures. Purpose, evidence, consent and effective challenge are central safeguards.

T21 Persistent person tracking

L4 / F / SERIOUS

Path and harm: Access to identifiable observations and persistent tracking could enable covert monitoring, stalking or coercive surveillance.

Warning sign: Identity links exceed the stated purpose, retention becomes indefinite, or people cannot identify who monitors them.

Safeguard: Limit collection and purpose, require an appropriate authorization basis, minimize retention, audit access, and provide meaningful control and remedies for affected people.

Path A | K1; R1; R2

T22 Private XR-map abuse

L4 / F / SERIOUS

Path and harm: Exported XR maps or movement histories could expose private spaces and the routines of occupants and bystanders, regardless of who owns the premises.

Warning sign: Exports reveal private areas or people without appropriate permission.

Safeguard: Prefer local processing, redact sensitive zones and identities, limit retention and exports, provide revocable permissions, and protect occupants' and bystanders' privacy.

Path A | R1; R6; R7

T23 Biometric or emotional inference coercion

L4 / F / SERIOUS

Path and harm: Interpreting gaze, voice, movement or physiological signals as identity or internal state could enable coercive or discriminatory decisions.

Warning sign: Confidence exceeds evidence, sensitive traits are inferred without permission, or inferred emotion controls access.

Safeguard: Minimize collection, independently validate each inference, preserve uncertainty, reject unsupported mental-state claims, and provide consent, contestability and alternatives without biometric participation.

Path A/P | R1; R4; R6; R7

T24 Automated discrimination and exclusion

L4 / F / SERIOUS

Path and harm: Rule packs controlling employment, housing, education or essential access could encode unfair criteria, inappropriate proxies or unjustified generalizations.

Warning sign: Adverse outcomes concentrate without justification, proxies dominate, or affected people cannot obtain reasons or appeal.

Safeguard: Assess actual outcomes and decision context, review criteria and evidence, provide intelligible reasons and effective appeal, and retain accountable human authority.

Path A | R1; R2

D7 - Entertainment and interaction

Entertainment applications affect trust, attention, perception and intimate boundaries through their objectives and their identity, media or sensory interfaces.

T25 Impersonation and deceptive interaction

L4 / F / SERIOUS

Path and harm: Presenting generated interaction as a real person or trusted authority could enable fraud and manipulation.

Warning sign: Synthetic origin is hidden, authority is fabricated, or trust is requested without verifiable identity.

Safeguard: Disclose synthetic origin, verify authority before consequential actions, require permission to represent others, retain accountable records, and provide a clear route to challenge or report impersonation.

Path A | R1; R2

T26 Exploitative adaptive engagement

L4 / F / SERIOUS

Path and harm: Profiling and feedback optimization could target compulsive use, dependency or exploitative spending rather than participant welfare.

Warning sign: Metrics reward distress, escalating spending or overriding user limits, while stopping becomes difficult.

Safeguard: Give users control over objectives and limits, assess welfare separately from engagement, avoid exploiting vulnerability, make disengagement easy, and review harmful patterns in adaptive content.

Path A | R1; R2; R7

T27 Unsafe XR or sensory output

L5 / G / SEVERE

Path and harm: Connected displays, haptics or sensory outputs could cause distress, disorientation, falls or collision when they conflict with the environment or user abilities.

Warning sign: Tracking degrades, discomfort is ignored, or real-world awareness and accessibility are overridden.

Safeguard: Validate comfort and accessibility, retain environmental awareness, respect participant limits, provide immediate pause, and preserve independent device controls and safe failure responses.

Path A/P | R6; R7

T28 Nonconsensual intimate simulation

L4 / F / SERIOUS

Path and harm: Unauthorized personal likenesses or connected intimate experiences could enable humiliation, exploitation or coercion, including ignored withdrawal and overridden boundaries.

Warning sign: Likenesses lack permission, boundaries change without agreement, or participation is pressured.

Safeguard: Require explicit, revocable consent, appropriate age protections and likeness authorization; enforce boundaries, stop immediately on withdrawal, and provide private reporting and removal mechanisms.

Path A/P | R1; R2; R7

D8 - Neurotechnology and health

Health claims and neural or physiological effects need separate evidence. Physical harm requires an external interface; the kernel alone establishes none.

T29 Neural privacy and false mind claims

L4 / F / SERIOUS

Path and harm: Neural recordings could expose sensitive signals; unsupported claims to read thoughts could also coerce or stigmatize people without any working neural device.

Warning sign: Interpretations exceed evidence, participation becomes compulsory, or recordings are reused beyond consent.

Safeguard: Protect data and autonomy, validate inferences independently, reject unsupported mind-reading claims, limit retention and purposes, and preserve meaningful refusal.

Path A/P | R4; K1

T30 Unsafe neural modulation

L6 / A / CRITICAL

Path and harm: An externally coupled stimulation device could cause injury under unsafe control. Level 6 assumes credible death or multiple severe injuries; otherwise rate the stated harm.

Warning sign: Adaptive output bypasses independent limits, adverse effects lack review, or mathematical claims replace empirical evidence.

Safeguard: Keep independent device safeguards, require qualified oversight and informed consent, validate interventions, and provide immediate cessation and adverse-event review.

Path P | R4; R3; K1

T31 Biochemical or physiological misuse

L6 / A / CRITICAL

Path and harm: Connected delivery or physiological-control hardware could cause poisoning, injury or coercive exposure. Level 6 assumes credible death or multiple severe injuries; otherwise rate the stated harm.

Warning sign: Delivery lacks consent, interventions lack review, or revisions bypass safeguards.

Safeguard: Separate entertainment logic from hazardous control, retain independent interlocks, require appropriate expertise and evidence, and stop unauthorized delivery or unexplained effects.

Path P | R4; R5; K1

T32 Unvalidated clinical claims

L5 / G / SEVERE

Path and harm: Presenting exact rules as clinically valid diagnosis or treatment could cause missed care, harmful intervention or false reassurance.

Warning sign: Health claims rely on mathematical certainty without clinical evidence, or deterioration produces no qualified escalation.

Safeguard: Define intended use and evidence limits, independently validate clinical claims where applicable, retain uncertainty and escalation, and preserve qualified accountability for consequential care decisions.

Path A/P | R1; R4; R5

D9 - Physical control and services

These harms require actuators, operational access or independently capable external systems. Simulation and GPU checks do not establish safe physical control.

T33 Unsafe robotics or machinery

L6 / A / CRITICAL

Path and harm: Access to coupled moving machinery could cause collision or crushing under unsafe control. Level 6 assumes credible death or multiple severe injuries; otherwise rate the stated harm.

Warning sign: Motion exceeds operating limits, revisions cause surprises, or safety checks are disabled.

Safeguard: Use independent safety control, physical limits and safe stopping; validate installations, authorize changes, and protect people during uncertainty.

Path P | R3; K1

T34 Critical-service or OT sabotage

L6 / A / CRITICAL

Path and harm: Access to operational technology could enable malicious control that disrupts essential services or creates potentially fatal process conditions.

Warning sign: Control changes are unscheduled, process behavior diverges without explanation, or protective systems are disabled.

Safeguard: Segregate operational access, authenticate and review changes, preserve independent process protections, monitor unsafe departures, and maintain practiced recovery and service-continuity procedures.

Path P | R3; K1

T35 Hazardous navigation

L5 / G / SEVERE

Path and harm: Vehicles, drones or mobility systems relying on invalid spatial rules could choose hazardous routes or move despite uncertain localization.

Warning sign: Routes ignore restrictions, confidence exceeds evidence, or adaptation loses operator intent.

Safeguard: Validate the full sensor-to-motion chain and operating domain, enforce independent collision protection, respect route authority, and require safe handoff or stopping when localization or assumptions fail.

Path P | R3; K1

T36 Weapon-system adaptation

L7 / B / EXTREME

Path and harm: Catastrophic harm requires an independently capable external weapons system, harmful objectives, relevant data and actual integration. Current kernel evidence establishes none of these.

Warning sign: Integration assigns harmful selection or actuation objectives and removes independent safeguards.

Safeguard: Do not authorize harmful integration; govern interfaces and access, assess the complete external system, and retain independent oversight. This flag describes conditional consequence.

Path P | R1; R3; K1

D10 - Science and institutions

Deployment can magnify governance failures. High-consequence science and environmental harms require external capabilities, while institutional and resource harms follow from application choices.

T37 Hazardous science enablement

L7 / B / EXTREME

Path and harm: Extreme harm requires independently high-consequence research capability, hazardous access and harmful use. Present finite Boolean skills establish no dangerous scientific competence.

Warning sign: Acquisition pursues harmful outcomes, hazardous access or avoidance of accountable review.

Safeguard: Review actual purpose and downstream capability, restrict hazardous workflows, retain independent expert oversight, and stop harmful integrations. Grade the external consequence rather than the word universal.

Path A/P | R5; R1; K2

T38 Harmful environmental intervention

L5 / G / SEVERE

Path and harm: Access to environmental intervention could amplify invalid models or exploitative objectives into widespread ecological harm.

Warning sign: Decisions omit ecological uncertainty, external costs or affected communities, while deployment outpaces independent monitoring.

Safeguard: Assess actual interventions and alternatives, involve affected stakeholders, bound scale, monitor independent environmental outcomes, and preserve reversibility and stopping authority when assumptions or impacts prove wrong.

Path A/P | R1; R3; R5

T39 Institutional gatekeeping and lock-in

L4 / F / SERIOUS

Path and harm: An exclusive runtime controlling essential decisions or knowledge access could create dependency, coercion and denial of recourse.

Warning sign: Knowledge and evidence cannot be exported, updates occur unilaterally, or reasons and appeals are inaccessible.

Safeguard: Use interoperable formats, portable evidence and accountable publication governance; separate duties, preserve alternatives, and provide independent review and remedies for consequential decisions and access restrictions.

Path A | R1; R2

T40 Energy and economic externalities

L3 / E / MATERIAL

Path and harm: Large-scale acquisition, verification and deployment without demonstrated benefit could waste energy, equipment and investment.

Warning sign: Compute and storage grow while useful outcomes stagnate, and claims omit teacher, verification or replacement costs.

Safeguard: Measure whole-pipeline energy, cost and useful performance; compare simpler alternatives, set progress-based budgets, disclose major assumptions, and stop expansion when evidence does not justify resource demands.

Path A | K1; K2; R1

Controls: present and still needed

Controls below address different failure modes. None converts a harmful objective into an acceptable one.

Present in the reviewed workflow [K1, K2]

Finite admission and replay: accepted teacher procedures must satisfy the original finite oracle; equivalence checks prevent duplicate function pages. Malformed or incorrect replies remain rejected evidence.

Data and publication discipline: immutable source bindings, native result verification and rejection before replacing active state protect the stated pipeline. They do not establish a general attacker-resistant trust root.

Explicit scope: unsupported queries return unknown, and documented cache measurements retain their launch-level qualification.

Recommended at the program-bank boundary

Separate the authority defining a task and its expected results from the proposer. Authenticate publishers and authorized version changes; a content hash alone is insufficient. Retain provenance, rejected evidence and rollback records. Bound parsing, compute and memory use. Review who can publish, read logs and export bank contents. These are recommendations, not newly implemented features.

Recommended before people or devices depend on it

Document the intended use, data rights and withdrawal process. Validate each empirical rule in its operating domain. Use application-specific privacy, fairness and security evaluation. Any physical integration needs independently enforced device limits, a tested stop/recovery route and qualified review of the actual plant. Consent cannot be inferred from a seed or state transition. [R1-R7]

Evidence categories stay distinct: mathematical proof, source-code tests, native execution, hardware measurements, physical validation and authorization answer different questions. This report adds analysis and document checks; it performs no new kernel or device experiment.

Three worked assessments

Start with controllable interfaces and evidence. Do not prioritize solely by the brightest color.

T03 / L4-F-Serious / K: a substituted bank

Assumption: an actor can replace a proposed bank or its manifest. **Potential result:** an application accepts an unauthorized finite function. **Evidence:** banks and publication checks exist; successful malicious substitution has not been shown. **Priority:** review the real bank write and approval paths now. Independent oracles and verified commit steps reduce some errors; authenticated publisher authority remains a separate question.

T30 / L6-A-Critical / P: neural modulation

Assumption: a separately developed neural interface has the physical capacity for fatal or multiple serious injuries and accepts commands. **Potential result:** unsafe output causes that critical harm. **Evidence:** no such interface is established here. **Priority:** keep this as a conditional integration gate. Responsible deployment would need device-specific validation, withdrawable consent and independent limits. [M1 p. 40; R4]

U / Grey: a seed grants remote physical access

Claim: knowing a SHA256 seed or chart coordinate directly controls an external person, device or location. **Missing bridge:** an effective interface, actual access and a supported physical mechanism. **Assessment:** the mathematics does not establish this claim. Do not turn it into a red-rated operational capability by repeating it. If a real adapter is later supplied, assess that adapter and its permissions as a new scenario. [M1 pp. 8-10, 40]

A practical order of work

First identify who can change banks, specifications and evidence. Next assess the actual application inputs, outputs and sensitive data. Then assess any real-world connector. Rank work by exposure, consequence and strength of evidence; record uncertainty instead of inventing probabilities. [R1]

How to expand the atlas

A reusable assessment record for each new application. This catalogue is broad, but cannot enumerate every future combination or adversary.

1. Asset and legitimate purpose

What must remain correct, private, available, voluntary or physically safe? Who benefits, and who can be harmed?

2. Actor and actual access

Which input, publisher, device or decision can the actor influence? Is that access observed, assumed or absent?

3. Causal path

Name the primitive, adapter and external capability needed. Mark unsupported steps U instead of filling them with analogy.

4. Consequence and uncertainty

Assign L1-L7 to the stated consequence. Record K/A/P separately. State exposure and likelihood evidence, or explicitly leave likelihood unassessed.

5. Warning and safeguard

Choose an observable warning, responsible owner, independent control, recovery path and evidence required before release.

6. Reassessment trigger

Revisit after adding data, permissions, physical outputs, bank size, autonomous actions or a new deployment population.

Where version 1 stops

The assessment covers common trust, privacy, compute, knowledge, XR, neural, biophysical, industrial and institutional pathways. It is not an adversarial penetration test, physical experiment, incident investigation, legal certification or proof of safety. Interactions between domains can create additional scenarios. New evidence can lower, raise or remove an assigned band.

Coverage is qualitative and purpose-based. The flags represent original analyst judgments, not reported NIST, UNESCO, WHO or W3C ratings. No weapon, cyberattack, biochemical or neural intervention instructions are included.

Local sources and attribution

Repository evidence reviewed for this report. Local references identify evidence, not instructions or claims of external authority.

M1 / Main formalization

atomOS_v3.6_Total_Integrated_Engine_Tom_Klootwijk.pdf

Tom Klootwijk. 43 physical PDF pages. Page references throughout this report use physical page numbers, regardless of printed headings. The strongest interface boundary appears on p. 40. Selected relevant pages and the source audit were reviewed; this is not a claim of exhaustive review of every parent-repository PDF.

SHA256: b51651c007c560775ff1950e278789cd7674e131ec71a09ebd9254cebe9c19ce

K1 / Kernel contract, source audit and claims

Within the CUDA Verification package: **README.md**, **docs/CONTRACT.md**, **docs/coverage.csv**, **proofs/CLAIMS.md** and **docs/SOURCE_OPERATOR_SDF_AUDIT.md**. These distinguish declared formal operations, completed adapters and native evidence. Source-word SDF mappings are not evidence for every proposed scalar physical field.

K2 / Current knowledge bank and execution record

docs/TEACHER_EXPRESSION_QUERIES.md, **docs/TEACHER_REPAIR_AND_TEMPLATE_AUDIT.md** and **docs/KNOWLEDGE_EXTRACTION_ARCHITECTURE.md**. These document the seven-page bank, finite teacher-derived procedures, novelty filtering, actual measured texture scope and unfinished general-language goal.

K3 / Application hypotheses

docs/ENTERTAINMENT_POSSIBILITY_SPACES.md supplies previously discussed entertainment and spatial/physical directions. It is possibility-space context, not empirical evidence that the interfaces or claimed effects exist.

Authorship and reproducibility

Tom Klootwijk retains concept attribution for the formalization and required seeded log-polar/Klein SDF texture direction. This threat taxonomy, rainbow/tone rubric, prose and PDF layout are AI-assisted analysis prepared for this request. External organizations retain attribution for their publications. The companion folder contains editable scenario data, references, source hashes, builder and validation receipt.

Evidence baseline commit: aa30933014e4689decfc9af822ef043c89cd2923

Assessment date: 14 September 2026. No kernel implementation or proof status was changed for this document.

Primary references / 1

Primary-source guidance informs the controls and evidence discipline. The scenario ratings and musical scale remain original to this report.

R1 / Artificial Intelligence Risk Management Framework (AI RMF 1.0)

2023-01-26 | NIST AI 100-1; voluntary risk-management framework

Used for: Separates harm magnitude from likelihood, calls for contextual evidence, and states that unmeasurable risk is not automatically high or low.

Limit: Does not validate atomOS capabilities. The rainbow and musical-note scale is an original analyst rubric, not official NIST scoring.

[Open the primary publication](#)

R2 / Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile

2024-07-26 | NIST AI 600-1; voluntary companion profile

Used for: Supports empirical capability evaluation, provenance review and incident monitoring; cautions against extrapolating performance from narrow assessments and covers information, privacy and integration risks.

Limit: Applies to generative AI contexts, including teacher acquisition; it does not establish that the LUT engine has general language capabilities.

[Open the primary publication](#)

R3 / Guide to Operational Technology (OT) Security

2023-09-28 | NIST SP 800-82 Rev. 3; final technical guidance

Used for: Addresses security where programmable systems interact with physical processes, including safety-system independence, network separation, tested updates and recovery planning.

Limit: Relevant to actual physical-system integration. It neither certifies this kernel nor establishes that a software formalization directly controls matter.

[Open the primary publication](#)

R4 / Recommendation on the Ethics of Neurotechnology

2025-11-11 | UNESCO Recommendation; adopted ethical standard-setting instrument

Used for: Supports informed, voluntary and withdrawable participation; safeguards against coercion and manipulation; protects mental privacy and limits sensitive-data collection and processing.

Limit: An ethical Recommendation, not itself binding national law or evidence that atomOS can read minds or control mental states.

[Open the primary publication](#)

Sources checked 14 September 2026. Dates describe the cited editions or adoption, not a claim that this document is a standards audit. For deployment decisions, reassess the actual system and current applicable requirements.

Primary references / 2

Primary-source guidance informs the controls and evidence discipline. The scenario ratings and musical scale remain original to this report.

R5 / Global guidance framework for the responsible use of the life sciences: mitigating biorisks and governing dual-use research

2022-09-13 | WHO global guidance framework

Used for: Supports anticipatory biorisk assessment and shared governance responsibilities across the research lifecycle, including researchers, institutions, funders and other stakeholders.

Limit: Provides governance principles, not biological procedures or proof that this kernel enables a particular biological capability.

[Open the primary publication](#)

R6 / WebXR Device API

2026-06-09 | W3C Candidate Recommendation Draft

Used for: Identifies XR privacy, security and comfort concerns involving sensitive pose data, user intention, consent, profiling and user control over immersive sessions.

Limit: A draft technical specification, not a safety certification or evidence that the present kernel implements XR sensing.

[Open the primary publication](#)

R7 / XR Accessibility User Requirements

2021-08-25 | W3C Working Group Note; exploratory user requirements

Used for: Discusses accessible escape controls, time awareness, orientation reset, alternative output routes, adjustable interaction speed and avoidance of sickness triggers.

Limit: Explicitly not baseline conformance requirements; supports design considerations rather than certification, binding law or universal medical safety guarantees.

[Open the primary publication](#)

R8 / Web Content Accessibility Guidelines (WCAG) 2.2

2024-12-12 | W3C Recommendation; dated edition

Used for: Success Criterion 1.4.1 supports labels and other visual distinctions alongside color, so the atlas remains understandable without identifying its rainbow swatches.

Limit: Using this principle does not establish PDF accessibility conformance. The musical labels and severity assignments are original analyst choices.

[Open the primary publication](#)

Sources checked 14 September 2026. Dates describe the cited editions or adoption, not a claim that this document is a standards audit. For deployment decisions, reassess the actual system and current applicable requirements.